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(54) **SYSTEM AND METHOD FOR SURFACING CYBER-SECURITY THREATS WITH A SELF-LEARNING RECOMMENDATION ENGINE**

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CPC **H04L 63/1466** (2013.01); **G06F 11/327** (2013.01); **G06F 18/24** (2023.01); **G06N 20/00** (2019.01)

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(56) **References Cited**

U.S. PATENT DOCUMENTS

4,292,580 A 9/1981 Ott et al.
5,175,732 A 12/1992 Hendel et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 112567367 3/2019
EP 2990984 3/2016
(Continued)

OTHER PUBLICATIONS

Abdullah et al., "Visualizing Network Data for Intrusion Detection.", 2005 Institute of Electrical and Electronics Engineers Workshop on Information Assurance and Security, United States Military Academy, West Point, New York, United States, pp. 100-108.

(Continued)

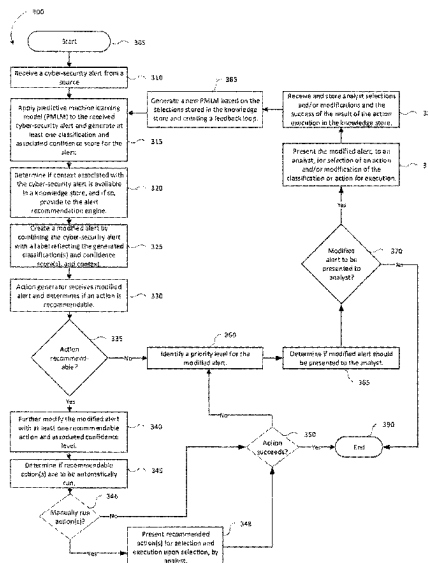
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(57) **ABSTRACT**

Techniques for performing cyber-security alert analysis and prioritization according to machine learning employing a predictive model to implement a self-learning feedback loop. The system implements a method generating the predictive model associated with alert classifications and/or actions which automatically generated, or manually selected by cyber-security analysts. The predictive model is used to determine a priority for display to the cyber-security analyst and to obtain the input of the cyber-security analyst to improve the predictive model. Thereby the method implements a self-learning feedback loop to receive cyber-security alerts and mitigate the cyberthreats represented in the cybersecurity alerts.

20 Claims, 3 Drawing Sheets



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(56)

References Cited

U.S. PATENT DOCUMENTS

8,464,340 B2	6/2013	Ahn et al.	9,282,109 B1	3/2016	Aziz et al.
8,479,174 B2	7/2013	Chiriack	9,292,686 B2	3/2016	Ismael et al.
8,479,276 B1	7/2013	Vaystikh et al.	9,294,501 B2	3/2016	Mesdaq et al.
8,479,291 B1	7/2013	Bodke	9,300,686 B2	3/2016	Pidathala et al.
8,510,827 B1	8/2013	Leake et al.	9,306,960 B1	4/2016	Aziz
8,510,828 B1	8/2013	Guo et al.	9,306,974 B1	4/2016	Aziz et al.
8,510,842 B2	8/2013	Amit et al.	9,311,479 B1	4/2016	Manni et al.
8,516,478 B1	8/2013	Edwards et al.	9,355,247 B1	5/2016	Thioux et al.
8,516,590 B1	8/2013	Ranadive et al.	9,356,944 B1	5/2016	Aziz
8,516,593 B2	8/2013	Aziz	9,363,280 B1	6/2016	Rivlin et al.
8,522,348 B2	8/2013	Chen et al.	9,367,681 B1	6/2016	Ismael et al.
8,528,086 B1	9/2013	Aziz	9,398,028 B1	7/2016	Karandikar et al.
8,533,824 B2	9/2013	Hutton et al.	9,413,781 B2	8/2016	Cunningham et al.
8,539,582 B1	9/2013	Aziz et al.	9,426,071 B1	8/2016	Caldejon et al.
8,549,638 B2	10/2013	Aziz	9,430,646 B1	8/2016	Mushtaq et al.
8,555,391 B1	10/2013	Demir et al.	9,432,389 B1	8/2016	Khalid et al.
8,561,177 B1	10/2013	Aziz et al.	9,438,613 B1	9/2016	Paithane et al.
8,566,476 B2	10/2013	Shiffer et al.	9,438,622 B1	9/2016	Staniford et al.
8,566,946 B1	10/2013	Aziz et al.	9,438,623 B1	9/2016	Thioux et al.
8,584,094 B2	11/2013	Dadhia et al.	9,459,901 B2	10/2016	Jung et al.
8,584,234 B1	11/2013	Sobel et al.	9,467,460 B1	10/2016	Otvagin et al.
8,584,239 B2	11/2013	Aziz et al.	9,483,644 B1	11/2016	Paithane et al.
8,595,834 B2	11/2013	Xie et al.	9,495,180 B2	11/2016	Ismael
8,627,476 B1	1/2014	Satish et al.	9,497,213 B2	11/2016	Thompson et al.
8,635,696 B1	1/2014	Aziz	9,507,935 B2	11/2016	Ismael et al.
8,682,054 B2	3/2014	Xue et al.	9,516,057 B2	12/2016	Aziz
8,682,812 B1	3/2014	Ranjan	9,519,782 B2	12/2016	Aziz et al.
8,689,333 B2	4/2014	Aziz	9,536,091 B2	1/2017	Paithane et al.
8,695,096 B1	4/2014	Zhang	9,537,972 B1	1/2017	Edwards et al.
8,713,631 B1	4/2014	Pavlyushchik	9,560,059 B1	1/2017	Islam
8,713,681 B2	4/2014	Silberman et al.	9,565,202 B1	2/2017	Kindlund et al.
8,726,392 B1	5/2014	McCorkendale et al.	9,591,015 B1	3/2017	Amin et al.
8,739,280 B2	5/2014	Chess et al.	9,591,020 B1	3/2017	Aziz
8,776,229 B1	7/2014	Aziz	9,594,904 B1	3/2017	Jain et al.
8,782,792 B1	7/2014	Bodke	9,594,905 B1	3/2017	Ismael et al.
8,789,172 B2	7/2014	Stolfo et al.	9,594,912 B1	3/2017	Thioux et al.
8,789,178 B2	7/2014	Kejriwal et al.	9,609,007 B1	3/2017	Rivlin et al.
8,793,278 B2	7/2014	Frazier et al.	9,626,509 B1	4/2017	Khalid et al.
8,793,787 B2	7/2014	Ismael et al.	9,628,498 B1	4/2017	Aziz et al.
8,805,947 B1	8/2014	Kuzkin et al.	9,628,507 B2	4/2017	Haq et al.
8,806,647 B1	8/2014	Daswani et al.	9,633,134 B2	4/2017	Ross
8,832,829 B2	9/2014	Manni et al.	9,635,039 B1	4/2017	Islam et al.
8,850,570 B1	9/2014	Ramzan	9,641,546 B1	5/2017	Manni et al.
8,850,571 B2	9/2014	Staniford et al.	9,654,485 B1	5/2017	Neumann
8,881,234 B2	11/2014	Narasimhan et al.	9,661,009 B1	5/2017	Karandikar et al.
8,881,271 B2	11/2014	Butler, II	9,661,018 B1	5/2017	Aziz
8,881,282 B1	11/2014	Aziz et al.	9,674,298 B1	6/2017	Edwards et al.
8,898,788 B1	11/2014	Aziz et al.	9,680,862 B2	6/2017	Ismael et al.
8,935,779 B2	1/2015	Manni et al.	9,690,606 B1	6/2017	Ha et al.
8,949,257 B2	2/2015	Shiffer et al.	9,690,933 B1	6/2017	Singh et al.
8,984,638 B1	3/2015	Aziz et al.	9,690,935 B2	6/2017	Shiffer et al.
8,990,939 B2	3/2015	Staniford et al.	9,690,936 B1	6/2017	Malik et al.
8,990,944 B1	3/2015	Singh et al.	9,736,179 B2	8/2017	Ismael
8,997,219 B2	3/2015	Staniford et al.	9,740,857 B2	8/2017	Ismael et al.
9,009,822 B1	4/2015	Ismael et al.	9,747,446 B1	8/2017	Pidathala et al.
9,009,823 B1	4/2015	Ismael et al.	9,756,074 B2	9/2017	Aziz et al.
9,027,135 B1	5/2015	Aziz	9,773,112 B1	9/2017	Rathor et al.
9,071,638 B1	6/2015	Aziz et al.	9,781,144 B1	10/2017	Otvagin et al.
9,104,867 B1	8/2015	Thioux et al.	9,787,700 B1	10/2017	Amin et al.
9,106,630 B2	8/2015	Frazier et al.	9,787,706 B1	10/2017	Otvagin et al.
9,106,694 B2	8/2015	Aziz et al.	9,792,196 B1	10/2017	Ismael et al.
9,118,715 B2	8/2015	Staniford et al.	9,824,209 B1	11/2017	Ismael et al.
9,159,035 B1	10/2015	Ismael et al.	9,824,211 B2	11/2017	Wilson
9,171,160 B2	10/2015	Vincent et al.	9,824,216 B1	11/2017	Khalid et al.
9,176,843 B1	11/2015	Ismael et al.	9,825,976 B1	11/2017	Gomez et al.
9,189,627 B1	11/2015	Islam	9,825,989 B1	11/2017	Mehra et al.
9,195,829 B1	11/2015	Goradia et al.	9,838,408 B1	12/2017	Karandikar et al.
9,197,664 B1	11/2015	Aziz et al.	9,838,411 B1	12/2017	Aziz
9,223,972 B1	12/2015	Vincent et al.	9,838,416 B1	12/2017	Aziz
9,225,740 B1	12/2015	Ismael et al.	9,838,417 B1	12/2017	Khalid et al.
9,241,010 B1	1/2016	Bennett et al.	9,846,776 B1	12/2017	Paithane et al.
9,251,343 B1	2/2016	Vincent et al.	9,870,298 B2 *	1/2018	Jackson G06F 11/3433
9,262,635 B2	2/2016	Paithane et al.	9,876,701 B1	1/2018	Caldejon et al.
9,268,936 B2	2/2016	Butler	9,888,016 B1	2/2018	Amin et al.
9,275,229 B2	3/2016	LeMasters	9,888,019 B1	2/2018	Pidathala et al.
			9,910,988 B1	3/2018	Vincent et al.
			9,911,319 B2 *	3/2018	Malhotra G08B 29/185
			9,912,644 B2	3/2018	Cunningham
			9,912,681 B1	3/2018	Ismael et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,912,684	B1	3/2018	Aziz et al.	2003/0191957	A1	10/2003	Hypponen et al.
9,912,691	B2	3/2018	Mesdaq et al.	2003/0200460	A1	10/2003	Morota et al.
9,912,698	B1	3/2018	Thioux et al.	2003/0212902	A1	11/2003	van der Made
9,916,440	B1	3/2018	Paithane et al.	2003/0229801	A1	12/2003	Kouznetsov et al.
9,921,978	B1	3/2018	Chan et al.	2003/0237000	A1	12/2003	Denton et al.
9,934,376	B1	4/2018	Ismael	2004/0003323	A1	1/2004	Bennett et al.
9,934,381	B1	4/2018	Kindlund et al.	2004/0006473	A1	1/2004	Mills et al.
9,946,568	B1	4/2018	Ismael et al.	2004/0015712	A1	1/2004	Szor
9,953,185	B2*	4/2018	Bendersky G06F 16/435	2004/0019832	A1	1/2004	Arnold et al.
9,954,890	B1	4/2018	Staniford et al.	2004/0047356	A1	3/2004	Bauer
9,973,531	B1	5/2018	Thioux	2004/0083408	A1	4/2004	Spiegel et al.
10,002,252	B2	6/2018	Ismael et al.	2004/0088581	A1	5/2004	Brawn et al.
10,019,338	B1	7/2018	Goradia et al.	2004/0093513	A1	5/2004	Cantrell et al.
10,019,573	B2	7/2018	Silberman et al.	2004/0111531	A1	6/2004	Staniford et al.
10,025,691	B1	7/2018	Ismael et al.	2004/0117478	A1	6/2004	Triulzi et al.
10,025,927	B1	7/2018	Khalid et al.	2004/0117624	A1	6/2004	Brandt et al.
10,027,689	B1	7/2018	Rathor et al.	2004/0128355	A1	7/2004	Chao et al.
10,027,690	B2	7/2018	Aziz et al.	2004/0165588	A1	8/2004	Pandya
10,027,696	B1	7/2018	Rivlin et al.	2004/0236963	A1	11/2004	Danford et al.
10,033,747	B1	7/2018	Paithane et al.	2004/0243349	A1	12/2004	Greifeneder et al.
10,033,748	B1	7/2018	Cunningham et al.	2004/0249911	A1	12/2004	Alkhatib et al.
10,033,753	B1	7/2018	Islam et al.	2004/0255161	A1	12/2004	Cavanaugh
10,033,759	B1	7/2018	Kabra et al.	2004/0268147	A1	12/2004	Wiederin et al.
10,050,998	B1	8/2018	Singh	2005/0005159	A1	1/2005	Oliphant
10,068,091	B1	9/2018	Aziz et al.	2005/0021740	A1	1/2005	Bar et al.
10,075,455	B2	9/2018	Zafar et al.	2005/0033960	A1	2/2005	Vialen et al.
10,083,302	B1	9/2018	Paithane et al.	2005/0033989	A1	2/2005	Paletta et al.
10,084,813	B2	9/2018	Evada	2005/0050148	A1	3/2005	Mohammadioun et al.
10,089,461	B1	10/2018	Ha et al.	2005/0086523	A1	4/2005	Zimmer et al.
10,097,573	B1	10/2018	Aziz	2005/0091513	A1	4/2005	Mitomo et al.
10,104,102	B1	10/2018	Neumann	2005/0091533	A1	4/2005	Omote et al.
10,108,446	B1	10/2018	Steinberg et al.	2005/0091652	A1	4/2005	Ross et al.
10,121,000	B1	11/2018	Rivlin et al.	2005/0108562	A1	5/2005	Khazan et al.
10,122,746	B1	11/2018	Manni et al.	2005/0114663	A1	5/2005	Cornell et al.
10,133,863	B2	11/2018	Bu et al.	2005/0125195	A1	6/2005	Brendel
10,133,866	B1	11/2018	Kumar et al.	2005/0149726	A1	7/2005	Joshi et al.
10,146,810	B2	12/2018	Shiffer et al.	2005/0157662	A1	7/2005	Bingham et al.
10,148,693	B2	12/2018	Singh et al.	2005/0183143	A1	8/2005	Anderholm et al.
10,165,000	B1	12/2018	Aziz et al.	2005/0201297	A1	9/2005	Deikari
10,169,585	B1	1/2019	Pilipenko et al.	2005/0210533	A1	9/2005	Copeland et al.
10,176,321	B2	1/2019	Abbasi et al.	2005/0238005	A1	10/2005	Chen et al.
10,181,029	B1	1/2019	Ismael et al.	2005/0240781	A1	10/2005	Gassoway
10,191,861	B1	1/2019	Steinberg et al.	2005/0262562	A1	11/2005	Gassoway
10,192,052	B1	1/2019	Singh et al.	2005/0265331	A1	12/2005	Stolfo
10,198,574	B1	2/2019	Thioux et al.	2005/0283839	A1	12/2005	Cowburn
10,200,384	B1	2/2019	Mushtaq et al.	2006/0010495	A1	1/2006	Cohen et al.
10,210,329	B1	2/2019	Malik et al.	2006/0015416	A1	1/2006	Hoffman et al.
10,216,927	B1	2/2019	Steinberg	2006/0015715	A1	1/2006	Anderson
10,218,740	B1	2/2019	Mesdaq et al.	2006/0015747	A1	1/2006	Van de Ven
10,242,185	B1	3/2019	Goradia	2006/0021029	A1	1/2006	Brickell et al.
10,567,402	B1	2/2020	Comeaux et al.	2006/0021054	A1	1/2006	Costa et al.
11,017,321	B1*	5/2021	Mishra G05B 23/0283	2006/0031476	A1	2/2006	Mathes et al.
11,637,862	B1*	4/2023	Sopan G06N 5/025	2006/0047665	A1	3/2006	Neil
				2006/0070130	A1	3/2006	Costea et al.
				2006/0075496	A1	4/2006	Carpenter et al.
				2006/0095968	A1	5/2006	Portolani et al.
				2006/0101516	A1	5/2006	Sudaharan et al.
				2006/0101517	A1	5/2006	Banzhaf et al.
				2006/0117385	A1	6/2006	Mester et al.
				2006/0123477	A1	6/2006	Raghavan et al.
				2006/0143709	A1	6/2006	Brooks et al.
				2006/0150249	A1	7/2006	Gassen et al.
				2006/0161983	A1	7/2006	Cothrell et al.
				2006/0161987	A1	7/2006	Levy-Yurista
				2006/0161989	A1	7/2006	Reshef et al.
				2006/0164199	A1	7/2006	Gilde et al.
				2006/0173992	A1	8/2006	Weber et al.
				2006/0179147	A1	8/2006	Tran et al.
				2006/0184632	A1	8/2006	Marino et al.
				2006/0191010	A1	8/2006	Benjamin
				2006/0221956	A1	10/2006	Narayan et al.
				2006/0236393	A1	10/2006	Kramer et al.
				2006/0242709	A1	10/2006	Seinfeld et al.
				2006/0248519	A1	11/2006	Jaeger et al.
				2006/0248582	A1	11/2006	Panjwani et al.
				2006/0251104	A1	11/2006	Koga
				2006/0288417	A1	12/2006	Bookbinder et al.
				2007/0006288	A1	1/2007	Mayfield et al.
				2007/0006313	A1	1/2007	Porras et al.
11,870,799	B1*	1/2024	Imrem H04L 63/102				
2001/0005889	A1	6/2001	Albrecht				
2001/0047326	A1	11/2001	Broadbent et al.				
2002/0018903	A1	2/2002	Kokubo et al.				
2002/0038430	A1	3/2002	Edwards et al.				
2002/0091819	A1	7/2002	Melchione et al.				
2002/0095607	A1	7/2002	Lin-Hendel				
2002/0116627	A1	8/2002	Tarbotton et al.				
2002/0144156	A1	10/2002	Copeland				
2002/0162015	A1	10/2002	Tang				
2002/0166063	A1	11/2002	Lachman et al.				
2002/0169952	A1	11/2002	DiSanto et al.				
2002/0184528	A1	12/2002	Shevenell et al.				
2002/0188887	A1	12/2002	Largman et al.				
2002/0194490	A1	12/2002	Halperin et al.				
2003/0021728	A1	1/2003	Sharpe et al.				
2003/0074578	A1	4/2003	Ford et al.				
2003/0084318	A1	5/2003	Schertz				
2003/0101381	A1	5/2003	Mateev et al.				
2003/0115483	A1	6/2003	Liang				
2003/0188190	A1	10/2003	Aaron et al.				

(56)

References Cited

U.S. PATENT DOCUMENTS

2007/0011174 A1	1/2007	Takaragi et al.	2009/0126016 A1	5/2009	Sobko et al.
2007/0016951 A1	1/2007	Piccard et al.	2009/0133125 A1	5/2009	Choi et al.
2007/0019286 A1	1/2007	Kikuchi	2009/0144823 A1	6/2009	Lamastra et al.
2007/0033645 A1	2/2007	Jones	2009/0158430 A1	6/2009	Borders
2007/0038943 A1	2/2007	Fitzgerald et al.	2009/0172815 A1	7/2009	Gu et al.
2007/0064689 A1	3/2007	Shin et al.	2009/0187992 A1	7/2009	Poston
2007/0074169 A1	3/2007	Chess et al.	2009/0193293 A1	7/2009	Stolfo et al.
2007/0094730 A1	4/2007	Bhikkaji et al.	2009/0198651 A1	8/2009	Shiffer et al.
2007/0101435 A1	5/2007	Konanka et al.	2009/0198670 A1	8/2009	Shiffer et al.
2007/0128855 A1	6/2007	Cho et al.	2009/0198689 A1	8/2009	Frazier et al.
2007/0142030 A1	6/2007	Sinha et al.	2009/0199274 A1	8/2009	Frazier et al.
2007/0143827 A1	6/2007	Nicodemus et al.	2009/0199296 A1	8/2009	Xie et al.
2007/0156895 A1	7/2007	Vuong	2009/0228233 A1	9/2009	Anderson et al.
2007/0157180 A1	7/2007	Tillmann et al.	2009/0241187 A1	9/2009	Troyansky
2007/0157306 A1	7/2007	Elrod et al.	2009/0241190 A1	9/2009	Todd et al.
2007/0168988 A1	7/2007	Eisner et al.	2009/0265692 A1	10/2009	Godefroid et al.
2007/0171824 A1	7/2007	Ruello et al.	2009/0271867 A1	10/2009	Zhang
2007/0174915 A1	7/2007	Gribble et al.	2009/0300415 A1	12/2009	Zhang et al.
2007/0192500 A1	8/2007	Lum	2009/0300761 A1	12/2009	Park et al.
2007/0192858 A1	8/2007	Lum	2009/0328185 A1	12/2009	Berg et al.
2007/0198275 A1	8/2007	Malden et al.	2009/0328221 A1	12/2009	Blumfield et al.
2007/0208822 A1	9/2007	Liang et al.	2010/0005146 A1	1/2010	Drako et al.
2007/0220607 A1	9/2007	Sprosts et al.	2010/0011205 A1	1/2010	McKenna
2007/0240218 A1	10/2007	Tuvell et al.	2010/0017546 A1	1/2010	Pao et al.
2007/0240219 A1	10/2007	Tuvell et al.	2010/0030996 A1	2/2010	Butler, II
2007/0240220 A1	10/2007	Tuvell et al.	2010/0031353 A1	2/2010	Thomas et al.
2007/0240222 A1	10/2007	Tuvell et al.	2010/0037314 A1	2/2010	Derdisci et al.
2007/0250930 A1	10/2007	Aziz et al.	2010/0043073 A1	2/2010	Kuwamura
2007/0256132 A2	11/2007	Oliphant	2010/0054278 A1	3/2010	Stolfo et al.
2007/0271446 A1	11/2007	Nakamura	2010/0058474 A1	3/2010	Hicks
2008/0005782 A1	1/2008	Aziz	2010/0064044 A1	3/2010	Nonoyama
2008/0018122 A1	1/2008	Zierler et al.	2010/0077481 A1	3/2010	Polyakov et al.
2008/0028463 A1	1/2008	Dagon et al.	2010/0083376 A1	4/2010	Pereira et al.
2008/0040710 A1	2/2008	Chiriac	2010/0115621 A1	5/2010	Staniford et al.
2008/0046781 A1	2/2008	Childs et al.	2010/0132038 A1	5/2010	Zaitsev
2008/0066179 A1	3/2008	Liu	2010/0154056 A1	6/2010	Smith et al.
2008/0072326 A1	3/2008	Danford et al.	2010/0180344 A1	7/2010	Malyshev et al.
2008/0077793 A1	3/2008	Tan et al.	2010/0192223 A1	7/2010	Ismael et al.
2008/0080518 A1	4/2008	Hoefflin et al.	2010/0220863 A1	9/2010	Dupaquis et al.
2008/0086720 A1	4/2008	Lekel	2010/0235831 A1	9/2010	Dittmer
2008/0098476 A1	4/2008	Syversen	2010/0251104 A1	9/2010	Massand
2008/0120722 A1	5/2008	Sima et al.	2010/0281102 A1	11/2010	Chinta et al.
2008/0134178 A1	6/2008	Fitzgerald et al.	2010/0281541 A1	11/2010	Stolfo et al.
2008/0134334 A1	6/2008	Kim et al.	2010/0281542 A1	11/2010	Stolfo et al.
2008/0141376 A1	6/2008	Clausen et al.	2010/0287260 A1	11/2010	Peterson et al.
2008/0184367 A1	7/2008	McMillan et al.	2010/0299754 A1	11/2010	Amit et al.
2008/0184373 A1	7/2008	Traut et al.	2010/0306173 A1	12/2010	Frank
2008/0189787 A1	8/2008	Arnold et al.	2011/0004737 A1	1/2011	Greenebaum
2008/0201778 A1	8/2008	Guo et al.	2011/0025504 A1	2/2011	Lyon et al.
2008/0209557 A1	8/2008	Herley et al.	2011/0041179 A1	2/2011	Stahlberg
2008/0215742 A1	9/2008	Goldschmidt et al.	2011/0047594 A1	2/2011	Mahaffey et al.
2008/0222729 A1	9/2008	Chen et al.	2011/0047620 A1	2/2011	Mahaffey et al.
2008/0263665 A1	10/2008	Ma et al.	2011/0055907 A1	3/2011	Narasimhan et al.
2008/0295172 A1	11/2008	Bohacek	2011/0078794 A1	3/2011	Manni et al.
2008/0301810 A1	12/2008	Lehane	2011/0093951 A1	4/2011	Aziz
2008/0307524 A1	12/2008	Singh et al.	2011/0099620 A1	4/2011	Stavrou et al.
2008/0313738 A1	12/2008	Enderby	2011/0099633 A1	4/2011	Aziz
2008/0320594 A1	12/2008	Jiang	2011/0099635 A1	4/2011	Silberman et al.
2009/0003317 A1	1/2009	Kasralikar et al.	2011/0113231 A1	5/2011	Kaminsky
2009/0007100 A1	1/2009	Field et al.	2011/0145918 A1	6/2011	Jung et al.
2009/0013408 A1	1/2009	Schipka	2011/0145920 A1	6/2011	Mahaffey et al.
2009/0031423 A1	1/2009	Liu et al.	2011/0145934 A1	6/2011	Abramovici et al.
2009/0036111 A1	2/2009	Danford et al.	2011/0167493 A1	7/2011	Song et al.
2009/0037835 A1	2/2009	Goldman	2011/0167494 A1	7/2011	Bowen et al.
2009/0044024 A1	2/2009	Oberheide et al.	2011/0173213 A1	7/2011	Frazier et al.
2009/0044274 A1	2/2009	Budko et al.	2011/0173460 A1	7/2011	Ito et al.
2009/0064332 A1	3/2009	Porras et al.	2011/0219449 A1	9/2011	St. Neitzel et al.
2009/0077666 A1	3/2009	Chen et al.	2011/0219450 A1	9/2011	McDougal et al.
2009/0083369 A1	3/2009	Marmor	2011/0225624 A1	9/2011	Sawhney et al.
2009/0083855 A1	3/2009	Apap et al.	2011/0225655 A1	9/2011	Niemela et al.
2009/0089879 A1	4/2009	Wang et al.	2011/0247072 A1	10/2011	Staniford et al.
2009/0094697 A1	4/2009	Provos et al.	2011/0265182 A1	10/2011	Peinado et al.
2009/0113425 A1	4/2009	Ports et al.	2011/0289582 A1	11/2011	Kejriwal et al.
2009/0125976 A1	5/2009	Wassermann et al.	2011/0302587 A1	12/2011	Nishikawa et al.
2009/0126015 A1	5/2009	Monastyrsky et al.	2011/0307954 A1	12/2011	Melnik et al.
			2011/0307955 A1	12/2011	Kaplan et al.
			2011/0307956 A1	12/2011	Yermakov et al.
			2011/0314546 A1	12/2011	Aziz et al.
			2012/0023593 A1	1/2012	Puder et al.

(56)

References Cited**U.S. PATENT DOCUMENTS**

- 2012/0054869 A1 3/2012 Yen et al.
 2012/0066698 A1 3/2012 Yanoo
 2012/0079596 A1 3/2012 Thomas et al.
 2012/0084859 A1 4/2012 Radinsky et al.
 2012/0096553 A1 4/2012 Srivastava et al.
 2012/0110667 A1 5/2012 Zubrilin et al.
 2012/0117652 A1 5/2012 Manni et al.
 2012/0121154 A1 5/2012 Xue et al.
 2012/0124426 A1 5/2012 Maybee et al.
 2012/0174186 A1 7/2012 Aziz et al.
 2012/0174196 A1 7/2012 Bhogavilli et al.
 2012/0174218 A1 7/2012 McCoy et al.
 2012/0198279 A1 8/2012 Schroeder
 2012/0210423 A1 8/2012 Friedrichs et al.
 2012/0222121 A1 8/2012 Stanford et al.
 2012/0255015 A1 10/2012 Sahita et al.
 2012/0255017 A1 10/2012 Sallam
 2012/0260342 A1 10/2012 Dube et al.
 2012/0266244 A1 10/2012 Green et al.
 2012/0278886 A1 11/2012 Luna
 2012/0297489 A1 11/2012 Dequevy
 2012/0330801 A1 12/2012 McDougal et al.
 2012/0331553 A1 12/2012 Aziz et al.
 2013/0014259 A1 1/2013 Gribble et al.
 2013/0036472 A1 2/2013 Aziz
 2013/0047257 A1 2/2013 Aziz
 2013/0074185 A1 3/2013 McDougal et al.
 2013/0086684 A1 4/2013 Mohler
 2013/0097699 A1 4/2013 Balupari et al.
 2013/0097706 A1 4/2013 Titonis et al.
 2013/0111587 A1 5/2013 Goel et al.
 2013/0117852 A1 5/2013 Stute
 2013/0117855 A1 5/2013 Kim et al.
 2013/0139264 A1 5/2013 Brinkley et al.
 2013/0160125 A1 6/2013 Likhachev et al.
 2013/0160127 A1 6/2013 Jeong et al.
 2013/0160130 A1 6/2013 Mendelev et al.
 2013/0160131 A1 6/2013 Madou et al.
 2013/0167236 A1 6/2013 Sick
 2013/0174214 A1 7/2013 Duncan
 2013/0185789 A1 7/2013 Hagiwara et al.
 2013/0185795 A1 7/2013 Winn et al.
 2013/0185798 A1 7/2013 Saunders et al.
 2013/0191915 A1 7/2013 Antonakakis et al.
 2013/0196649 A1 8/2013 Padden et al.
 2013/0227691 A1 8/2013 Aziz et al.
 2013/0246370 A1 9/2013 Bartram et al.
 2013/0247186 A1 9/2013 LeMasters
 2013/0263260 A1 10/2013 Mahaffey et al.
 2013/0291109 A1 10/2013 Stanford et al.
 2013/0298243 A1 11/2013 Kumar et al.
 2013/0318038 A1 11/2013 Shiffer et al.
 2013/0318073 A1 11/2013 Shiffer et al.
 2013/0325791 A1 12/2013 Shiffer et al.
 2013/0325792 A1 12/2013 Shiffer et al.
 2013/0325871 A1 12/2013 Shiffer et al.
 2013/0325872 A1 12/2013 Shiffer et al.
 2014/0032875 A1 1/2014 Butler
 2014/0053260 A1 2/2014 Gupta et al.
 2014/0053261 A1 2/2014 Gupta et al.
 2014/0130158 A1 5/2014 Wang et al.
 2014/0137180 A1 5/2014 Lukacs et al.
 2014/0169762 A1 6/2014 Ryu
 2014/0179360 A1 6/2014 Jackson et al.
 2014/0181131 A1 6/2014 Ross
 2014/0189687 A1 7/2014 Jung et al.
 2014/0189866 A1 7/2014 Shiffer et al.
 2014/0189882 A1 7/2014 Jung et al.
 2014/0237600 A1 8/2014 Silberman et al.
 2014/0280245 A1 9/2014 Wilson
 2014/0283037 A1 9/2014 Sikorski et al.
 2014/0283063 A1 9/2014 Thompson et al.
 2014/0328204 A1 11/2014 Klotsche et al.
 2014/0337836 A1 11/2014 Ismael
 2014/0344926 A1 11/2014 Cunningham et al.
 2014/0351935 A1 11/2014 Shao et al.
 2014/0380473 A1 12/2014 Bu et al.
 2014/0380474 A1 12/2014 Paithane et al.
 2015/0007312 A1 1/2015 Pidathala et al.
 2015/0096022 A1 4/2015 Vincent et al.
 2015/0096023 A1 4/2015 Mesdaq et al.
 2015/0096024 A1 4/2015 Haq et al.
 2015/0096025 A1 4/2015 Ismael
 2015/0163242 A1 * 6/2015 Laidlaw H04L 63/1425
 726/22
 2015/0180886 A1 6/2015 Stanford et al.
 2015/0186645 A1 7/2015 Aziz et al.
 2015/0199513 A1 7/2015 Ismael et al.
 2015/0199531 A1 7/2015 Ismael et al.
 2015/0199532 A1 7/2015 Ismael et al.
 2015/0220735 A1 8/2015 Paithane et al.
 2015/0372980 A1 12/2015 Evada
 2016/0004869 A1 1/2016 Ismael et al.
 2016/0006756 A1 1/2016 Ismael et al.
 2016/0044000 A1 2/2016 Cunningham
 2016/0127393 A1 5/2016 Aziz et al.
 2016/0191547 A1 6/2016 Zafar et al.
 2016/0191550 A1 6/2016 Ismael et al.
 2016/0261612 A1 9/2016 Mesdaq et al.
 2016/0285914 A1 9/2016 Singh et al.
 2016/0301703 A1 10/2016 Aziz
 2016/0335110 A1 11/2016 Paithane et al.
 2017/0063901 A1 3/2017 Muddu et al.
 2017/0083703 A1 3/2017 Abbasi et al.
 2017/0243133 A1 * 8/2017 Zavesky G06N 20/00
 2018/0013770 A1 1/2018 Ismael
 2018/0048660 A1 2/2018 Paithane et al.
 2018/0121316 A1 5/2018 Ismael et al.
 2018/0150758 A1 5/2018 Niininen
 2018/0288077 A1 10/2018 Siddiqui et al.
 2018/0367561 A1 12/2018 Givental et al.
 2019/0020667 A1 1/2019 Parker
 2019/0236458 A1 * 8/2019 Taylor G06F 17/17
 2019/0260779 A1 * 8/2019 Bazalgette H04L 51/224
 2019/0334849 A1 10/2019 Bostick et al.
 2020/0145358 A1 5/2020 Yegorin et al.
 2020/0151326 A1 * 5/2020 Patrich G06F 21/554
 2020/0401696 A1 12/2020 Ringlein et al.
 2021/0058357 A1 2/2021 Baughman et al.
 2022/0174088 A1 * 6/2022 Zorlular H04L 41/22

FOREIGN PATENT DOCUMENTS

- GB 2439806 1/2008
 GB 2490431 10/2012
 GB 2520987 6/2015
 WO WO 0206928 1/2002
 WO WO 0223805 3/2002
 WO WO 2007117636 10/2007
 WO WO 2008041950 4/2008
 WO WO 2011084431 7/2011
 WO WO 2011112348 9/2011
 WO WO 2012075336 6/2012
 WO WO 2012145066 10/2012
 WO WO 2013067505 5/2013
 WO WO-2018126286 A1 * 7/2018 G06N 20/00

OTHER PUBLICATIONS

- Adetoye et al., "Network Intrusion Detection & Response System.", Adetoye, Sep. 2003.
 Apostolopoulos, "V-eM: A cluster of Virtual Machines for Robust, Detailed, and High-Performance Network Emulation.", Fourteenth Institute of Electrical and Electronics Engineers International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems, Sep. 11-14, 2006, pp. 117-126.
 Aura et al., "Scanning Electronic Documents for Personally Identifiable Information.", WPES '06: Fifth Association for Computing Machinery Workshop on Privacy in Electronic Society, Alexandria, Virginia, United States, Oct. 30, 2006, pp. 41-50.
 Baecher et al., "The Nepenthes Platform: An Efficient Approach to Collect Malware.", RAID '06: Ninth International Symposium on

(56)

References Cited**OTHER PUBLICATIONS**

Recent Advances in Intrusion Detection, Hamburg, Germany, Sep. 20-22, 2006, pp. 165-184.

Bayer et al., "Dynamic Analysis of Malicious Code.", *Journal in Computer Virology*, vol. 2, Nov. 2006, pp. 67-77.

Bouabalos, "Extracting Syslog Data Out of Raw PCAP Dumps.", [seclists.org](https://seclists.org/honeypots/2003/q2/319), retrieved on Aug. 8, 2023, Honeypots Mailing List Archives, <https://seclists.org/honeypots/2003/q2/319>, Jun. 5, 2003, pages.

Chaudet, et al., "Optimal Positioning of Active and Passive Monitoring Devices.", *CoNEXT '05: 2005 Association for Computing Machinery Conference on Emerging Network Experiment and Technology*, Toulouse, France, Oct. 24-27, 2005, pp. 71-82.

Chen et al., "When Virtual is Better Than Real.", *Eighth Workshop on Hot Topics in Operating Systems*, Elmau, Germany, May 20-22, 2001, 6 pages.

Christodorescu et al., "Mining Specifications of Malicious Behavior.", *ESEC-FSE '07: Sixth Joint Meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on the Foundations of Software Engineering*, Dubrovnik, Croatia, Sep. 3-7, 2007, pp. 5-14.

Cisco, "Intrusion Prevention for the Cisco ASA 5500-x Series.", *Data Sheet* 2012.

Cohen, "PyFlag—An Advanced Network Forensic Framework.", *Digital Investigation: The International Journal of Digital Forensics & Incident Response*, vol. 5, Sep. 2008, pp. S112-S120.

Costa et al., "Vigilante: End-to-End Containment of Internet Worms.", *Association for Computing Machinery Transactions on Computer Systems*, vol. 26, No. 4, Article 9, Dec. 2008, 68 pages.

Distler, "Malware Analysis: An Introduction.", *SANS Institute InfoSec Reading Room*, SANS Institute, Feb. 12, 2008, 67 pages.

Dunlap et al., "ReVirt: Enabling Intrusion Analysis through Virtual-Machine Logging and Replay.", *Fifth Symposium on Operating Systems Design and Implementation, OSDI '02 Fifth Symposium on Operating Systems Design and Implementation*, USENIX Association, Boston, Massachusetts, United States, Dec. 9-11, 2002, 14 pages.

FireEye, "Malware Analysis.", *Modern Malware Forensics*, FireEye Inc., 2010.

FireEye.com, "FireEye Malware Analysis & Exchange Network, Malware Protection System.", Feb. 2010, chrome-extension://efaidnbmnnnibpcapgleclefindmkaj/https://people.ucsc.edu/~warner/fireeye.pdf, retrieved on Sep. 19, 2023, 2 pages.

Goel et al., "Reconstructing System State for Intrusion Analysis.", *Apr. 2008 Special Interest Group on Operating Systems Review*, vol. 42, Issue 3, Apr. 2008, pp. 21-28.

Idika et al., "A Survey of Malware Detection Techniques.", *Department of Computer Science, Purdue University*, Feb. 2, 2007, 48 pages.

Keizer, "Microsoft's HoneyMonkeys Show Patching Windows Works.", Aug. 8, 2005, <https://www.informationweek.com/it-life/microsoft-s-honeymonkeys-show-patching-windows-works/>, retrieved Jun. 1, 2006, 4 pages.

Kim et al., "Autograph: Toward Automated, Distributed Worm Signature Detection.", *Thirteenth USENIX Security Symposium*, San Diego, California, United States, Aug. 9-13, 2004, pp. 271-286.

King et al., "Operating System Support for Virtual Machines.", *2003 USENIX Annual Technical Conference*, San Antonio, Texas, United States, Jun. 9-14, 2003, 15 pages.

Kreibich et al., "Honeycomb-Creating Intrusion Detection Signatures Using Honeypots.", *ACM SIGCOMM Computer Communication Review*, vol. 34, Issue 1, Jan. 2004, pp. 51-56.

Kristoff, "Botnets, Detection and Mitigation: DNS-Based Techniques.", *Security Day*, 2005, 23 pages.

Lastline Labs, "The Threat of Evasive Malware.", Feb. 25, 2013, Lastline Labs, pp. 1-8.

Li et al., "A VMM-Based System Call Interposition Framework for Program Monitoring.", *Institute of Electrical and Electronics Engineers Sixteenth International Conference on Parallel and Distributed Systems*, Shanghai, China, Dec. 2010, pp. 706-711.

Lindorfer et al., "Detecting Environment-Sensitive Malware.", *RAID 2011: Fourteenth International Symposium on Recent Advances in Intrusion Detection*, Menlo Park, California, United States, Sep. 20-21, 2011.

Marchette, "Computer Intrusion Detection and Network Monitoring: A Statistical Viewpoint.", 2001.

Moore et al., "Internet Quarantine: Requirements for Containing Self-Propagating Code.", *IEEE INFOCOM 2003: Twenty-second Annual Joint Conference of the Institute of Electrical and Electronics Engineers Computer and Communications Societies*, San Francisco, California, United States, Mar. 30-Apr. 3, 2003, pp. 1901-1910.

Morales et al., "Analyzing and Exploiting Network Behaviors of Malware.", *Security and Privacy in Communication Networks*. Springer Berlin Heidelberg, 2010, pp. 20-34.

Mori, "Detecting Unknown Computer Viruses.", 2004, Springer-Verlag Berlin Heidelberg.

Natvig, "Sandboxii: Internet.", *Virus Bulletin Conference* 2001, Sep. 2002, 18 pages.

NetBIOS Working Group, "Protocol Standard for a NetBIOS Service on a TCP/UDP Transport: Concepts and Methods.", *STD 19, RFC 1001*, Mar. 1987.

"Network Security NetDetector-Network Intrusion Forensic System (NIFS).", *NetDetector Whitepaper*, 2003, 11 pages.

Newsome et al., "Dynamic Taint Analysis for Automatic Detection, Analysis, and Signature Generation of Exploits on Commodity Software.", *Twelfth Annual Network and Distributed System Security Symposium (NDSS '05)*, San Diego, California, United States, Feb. 2-4, 2005, 17 pages.

Nojiri et al., "Cooperation Response Strategies for Large Scale Attack Mitigation.", *DARPA Information Survivability Conference and Exposition*, vol. 1, Apr. 22-24, 2003, pp. 293-302.

Oberheide et al., "CloudAV: N-Version Antivirus in the Network Cloud.", *Seventeenth USENIX Security Symposium* USENIX Security '08 Jul. 28-Aug. 1, 2008, San Jose, California, United States, 20 pages.

Ptacek et al., "Insertion, Evasion, and Denial of Service: Eluding Network Intrusion Detection.", *Secure Networks*, Jan. 1998, 63 pages.

Roundy et al., "Hybrid Analysis and Control of Malware.", *RAID 2010: Thirteenth International Symposium Recent Advances in Intrusion Detection*, Ottawa, Ontario, Canada, Sep. 15-17, 2010, pp. 317-338.

Sailer et al., "sHype: Secure Hypervisor Approach to Trusted Virtualized Systems.", *IBM Research Division*, Feb. 2, 2005, 13 pages.

Salah et al., "Using Cloud Computing to Implement a Security Overlay Network.", *Institute of Electrical and Electronics Engineers Security and Privacy Magazine*, vol. 11, No. 1, Jan. 2013, pp. 44-53.

Shinotsuka, "Malware Authors Using New Techniques to Evade Automated Threat Analysis Systems.", Oct. 26, 2012, <http://www.symantec.com/connect/blogs/>, pp. 1-4.

Silicon Defense, "Worm Containment in the Internal Network.", Mar. 2003, pp. 1-25.

Singh et al., "Automated Worm Fingerprinting.", *Proceedings of the ACM/USENIX Symposium on Operating System Design and Implementation*, San Francisco, California, United States, Dec. 2004.

Stevens, "Malicious PDF Documents Explained.", *Security & Privacy, Institute of Electrical and Electronics Engineers Security & Privacy*, vol. 9, No. 1, Jan.-Feb. 2011, pp. 80-82.

Yin et al., "Panorama: Capturing System-Wide Information Flow for Malware Detection and Analysis.", *CCS '07: Fourteenth ACM Conference on Computer and Communications Security*, Oct. 29-Nov. 2, 2007, Alexandria, Virginia, United States, pp. 116-127.

* cited by examiner

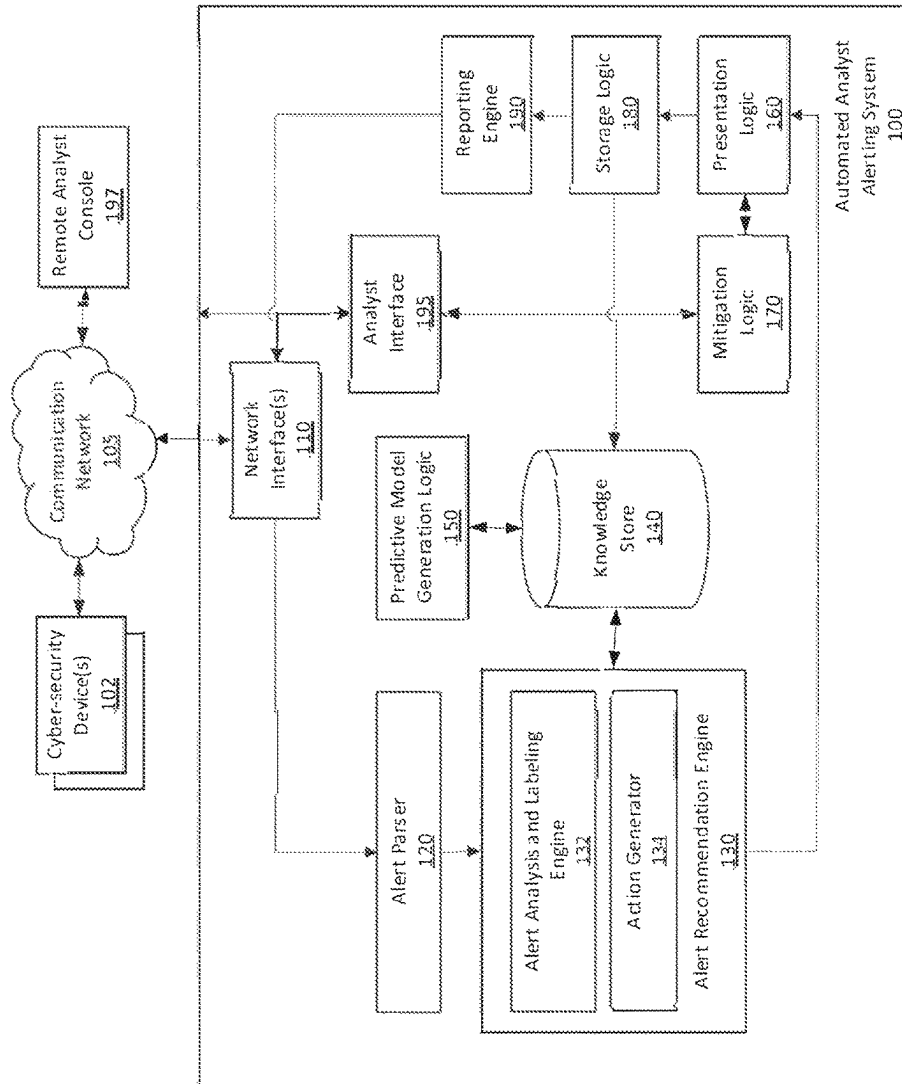


FIG. 1

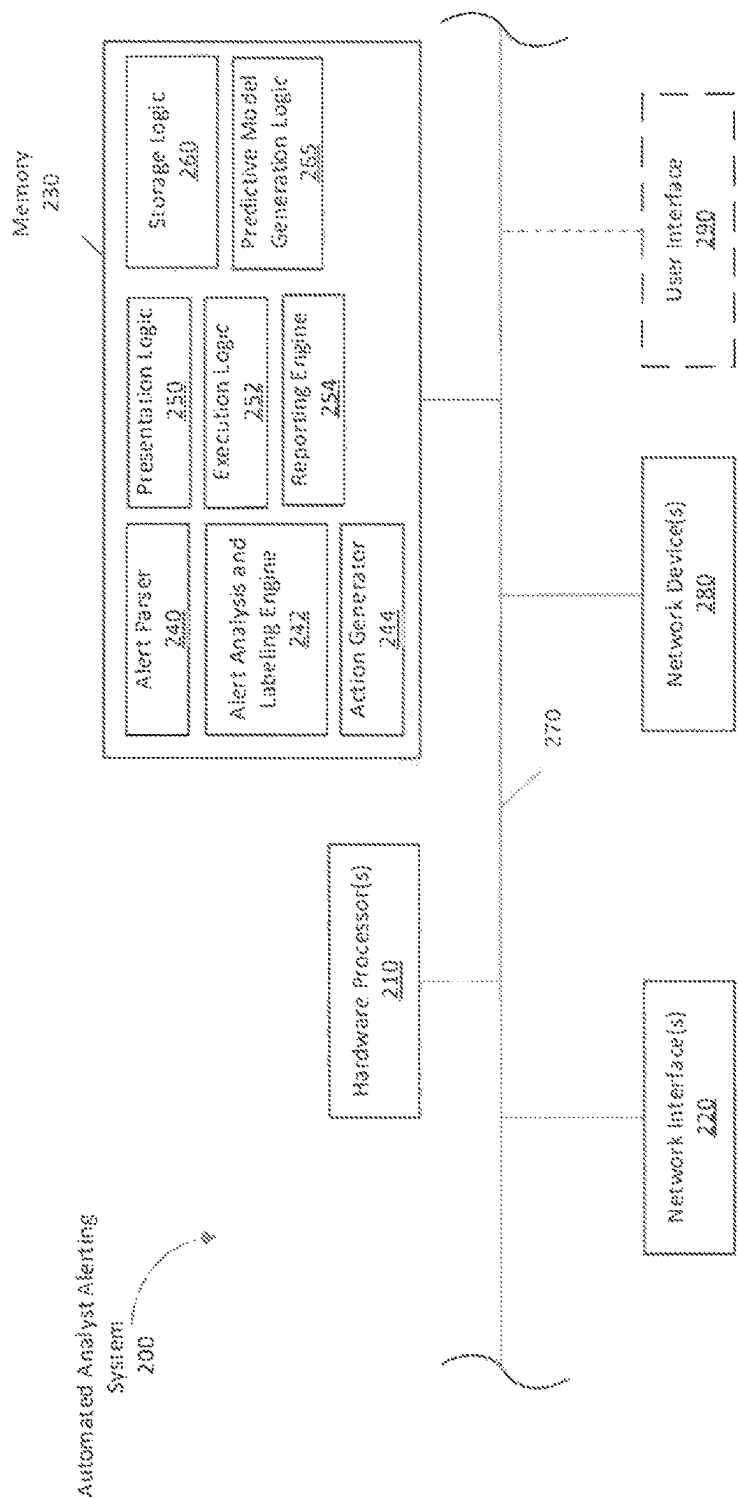


FIG. 2

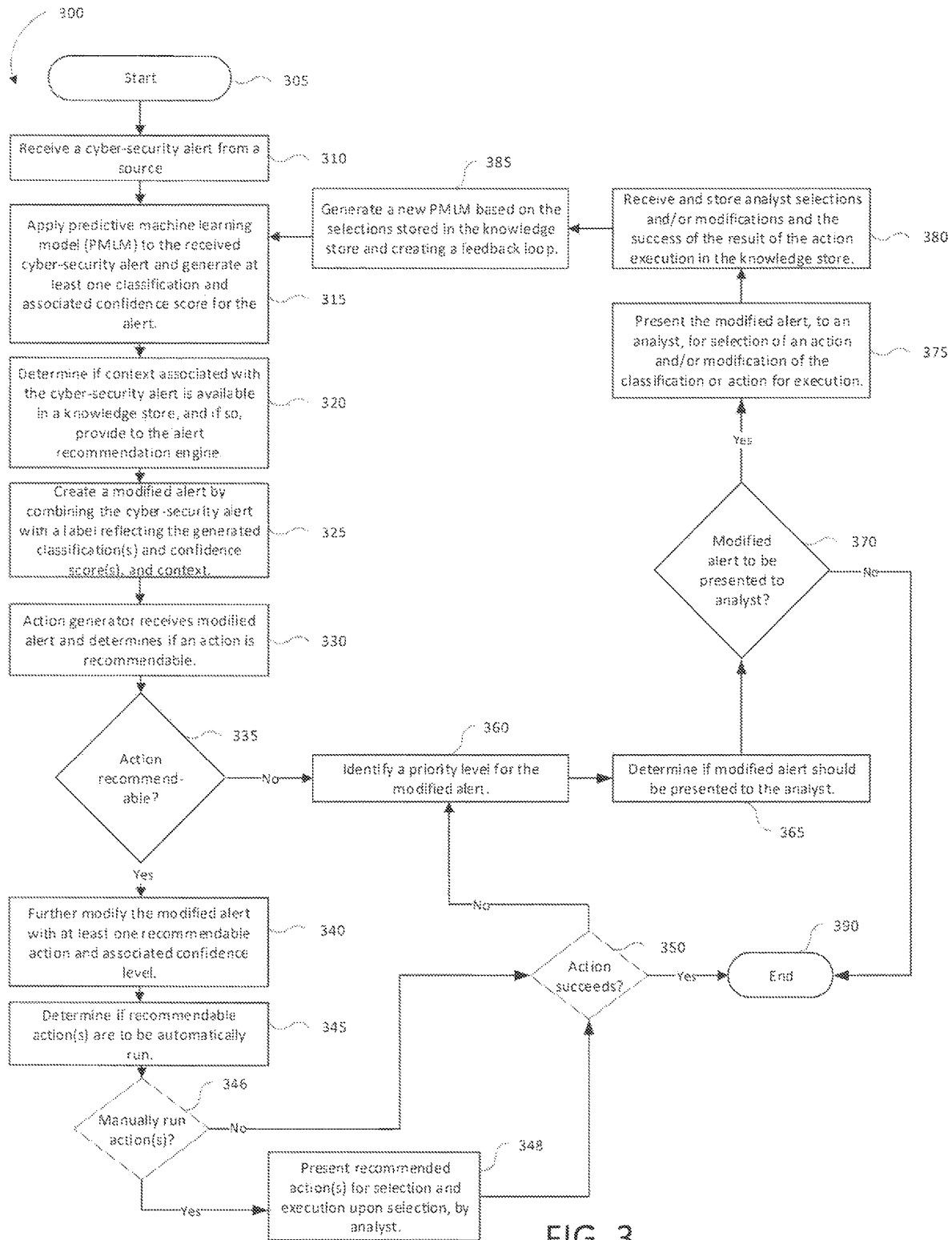


FIG. 3

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SYSTEM AND METHOD FOR SURFACING CYBER-SECURITY THREATS WITH A SELF-LEARNING RECOMMENDATION ENGINE

PRIORITY CLAIM

The present application is a continuation of U.S. application Ser. No. 16/588,967 having a filing date of Sep. 30, 2019, now U.S. Pat. No. 11,637,862. Applicant claims priority to and the benefit of each of such applications and incorporate all such applications herein by reference in its entirety.

FIELD OF THE INVENTION

The present disclosure relates, generally, to cyber-security and more specifically to techniques to facilitate the analysis and remediation of cyberattacks.

BACKGROUND

Cyber-security threats are a major risk to enterprises and individuals alike. Enterprises rely on security operations centers (“SOC”) and the analysts operating SOC’s, to identify, respond to, and mitigate the consequences of cyber-security threats targeting the enterprise’s systems. SOC analysts are inundated with cyber-security alerts received from a variety of cyber-security products deployed to protect an enterprise. To reduce the vast volume of alerts to be addressed by SOC analysts, some SOC’s filter alerts (e.g., for duplicates, known false positives, and low priority alerts, etc.) before they are presented to a SOC analyst.

BRIEF DESCRIPTION OF THE FIGURES

Embodiments of the disclosure are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings, in which like references indicate similar elements and in which:

FIG. 1 is an exemplary block diagram of an automated analyst alerting system communicatively coupled to one or more cyber-security devices via a communication network, in accordance with an embodiment of the invention.

FIG. 2 illustrates a logical representation of the automated analyst alerting system of FIG. 1.

FIG. 3 is an exemplary flowchart of the operations of the automated analyst alerting system of FIGS. 1 and 2.

DETAILED DESCRIPTION

The detailed description below, describes a technology wherein a cyber-security automated analyst alerting system receives one or more cyber-security alerts, the alerts are analyzed by an alert recommendation engine which automatically determines a recommended course of action related to the one or more received cyber-security alerts by application of a predictive machine learning model generated by a predictive machine learning logic (or predictive model generation logic). The predictive machine learning logic generates a machine learning model (or more simply, “model”), for use by the alert recommendation engine, in response to changes in a knowledge store. More specifically, to automatically determine a recommended course of action (i.e. a set of one or more instructions, or commands, issued by the described system to mitigate a cyber-security threat), the alert analysis and labeling engine generates a modified

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alert including at least one classification, classification confidence level, and contextual data for each alert according to the predictive machine learning model, to create a modified alert which are provided to the action generator. The action generator (or in some embodiments through the execution of an engine processing a separate action predictive machine learning model) determines a recommended course of action according to the predictive machine learning model and generates a signal through a reporting logic to present the modified alert for display to an analyst.

More specifically, the automated analyst alerting system (“AAAS”) is configured to receive an alert (the received alert is received from one or more alert-generating cyber-security devices), analyze the alert according to a model generated by a machine learning procedure applied to data in a knowledge store. The knowledge store includes data that associates previously detected alerts, cyber-security threats, and undesirable computing device configurations with one or more classifications as determined by a cyber-security analyst. Such classifications may include labels (e.g., “malicious”, “non-malicious”, “phishing”, “misconfiguration”, etc.) and a confidence level associated with the classification. For example, a received cyber-security alert received by the system and analyzed by the AAAS may classify the alert as “malicious” with a 17% confidence level, “non-malicious” with an 89% confidence level, and “misconfiguration” with a 91% confidence level. The classifications and their associated confidence levels are provided with the received alert, as well as with additional context related to the received alert, to create a modified alert and are provided to an action generator. The additional context may be based on prior selections of analysts, the prior selections stored in a knowledge store, and/or prior selections made by an expert system configured to make recommended actions based on associated received alerts. The knowledge store may be located locally and/or remotely via a network connection. In some embodiments the additional context may include information generated by the AAAS identifying a set of prior alerts (e.g., stored in the knowledge store) as being associated with a received alert and thereby identifying an advanced persistent cyber-security threat (i.e. a prolonged and targeted cyberattack in which an intruder may repeatedly attempt to gain access to a targeted network, computing device or user thereof). Based on the persistent cyber-security threat, the AAAS may modify the classifications and/or further classify the received alert as associated with the persistent cyber-security threat.

The predictive machine learning logic is configured to generate a predictive model based on data stored in the knowledge store. The data stored in the knowledge store may include the classifications associated with alerts that were previously received and classified (confirmed or reclassified) by cyber-security analysts. The knowledge store may also store mitigative actions selected by and/or input by a cyber-security analyst. The knowledge store may also be used to store meta-information associated with the success or failure of automated or manually selected mitigations and consequently create a self-learning feedback loop. The self-learning feedback loop surfaces classifications and actions for the cyber-security analysts.

The predictive machine learning logic may be co-located with the alert recommendation engine and/or remotely located. The predictive machine learning logic generates a predictive model according to conventional machine learning techniques (e.g., support vector machines, artificial neural networks, etc.) applied to the data stored in the knowledge store, in a process known as “training”. The

training system may include information extracted from received alerts and stored as data in the knowledge store. The information extracted from the received alert may include received alert message content as well as meta-information associated with the received alert (e.g., time of receipt, IP address of the source cyber-security device, etc.). The training system may also include information associated with the received alert (e.g., modifying a label associated with alert or associating a course of action with the alert) by the cyber-security analyst and stored in the knowledge store. Based on information stored in the knowledge store, the predictive machine learning logic may generate the predictive model which, when applied to a received alert, may be used to classify and determine one or more courses of action related to the received alert using machine learning.

In some embodiments, the generated predictive model may be used by one or more classifiers to determine a probability of the accuracy (i.e. confidence level) of a label for each alert. The classifiers may classify each alert based on a label as determined by an analyst and/or the alert recommendation engine according to the predictive model. In some embodiments, analysts may select from a pre-defined set of labels, whereas, in other embodiments, labeling may be done automatically. A classifier may generate a probability of association with a label relating to each received alert.

Upon receipt of new data in the knowledge store, or periodically or aperiodically to account for any such newly stored data, the predictive machine learning logic generates a new predictive model by analyzing the data to determine associative relationships. In some embodiments, the application of a predictive model to a received alert may generate one or more labels and/or courses of actions, each associated with a confidence level. The confidence levels are correlated with a likelihood of the alert being associated with the label and/or course of action. The newly generated predictive model may be based on additional data—e.g., verification of a prior classification (e.g., of a classification made by the alert recommendation engine and, in some embodiments confirmed by the analyst), newly associated courses of actions (i.e. mitigative actions responsive to a received alert), where the association may be made automatically or made or confirmed by an analyst, and/or new information associated with alert classification provided to the knowledge store via an update mechanism. The newly generated predictive model is applied to newly received alerts by the alert recommendation engine for classification, thereby creating a self-learning feedback loop. The classification is responsive to the labels resulting from application of the predictive model to the received alert.

The action generator receives the modified alerts and associated context information to determine a recommended course of action for presentation via the reporting logic. The action generator determines a recommended course of action based on the application of a predictive model generated by the predictive model generation logic. The received modified alerts are analyzed by the action generator to determine a priority for presentation to an analyst. To determine a priority associated with the modified alert, the action generator may analyze the confidence levels (e.g., associated with a course of action determined by application of the predictive model, associated with a classification label, etc.). The priority assigned to a received alert may be based, at least in part, on a numerical distance of the confidence level a threshold, such as, for example, an automated execution threshold. For example, if the confidence associated with an

action is 55% and the confidence threshold for automated execution of an action is 90%, the action generator may determine that the confidence associated with an action is too far from the threshold to be automatically actionable and should be displayed to an analyst and therefore given a higher priority for the analyst's attention. Similarly, if the confidence if the confidence associated with an action is 85% and the confidence threshold for automated execution of an action remains 90%, the action generator may determine that the confidence associated with an action is near the threshold, however, because it is not above the automatically actionable threshold, the received alert should be displayed to an analyst and therefore given a lesser priority than in the prior example. If a cyber-security threat or serious configuration issue requiring mitigation is detected (e.g., based on a classification and/or course of action), the action generator may determine whether the mitigation requires analyst attention (e.g., for selection) or if a recommended course of action may be automatically processed. To determine if analyst attention is required, the action generator determines if a course of action from the knowledge store and/or the expert system is applicable. A course of action is applicable if the action generator determines a level of correlation (i.e. confidence level) between a course of action and the modified alert exceeds a confidence threshold. If a course of action is automatically executed and fails to resolve the alert, the system may provide the modified alert associated with the failed action to the reporting logic for display to the analyst. If the action generator receives an alert associated with a persistent cyber-security threat, it may assign a priority to the modified alert and provide the priority to the presentation logic for display to an analyst. The action generator provides a further modified alert, the further modified alert combining the modified alert received by the action generator with the resulting course of actions, if applicable.

The further modified alert is provided to the presentation logic for layout composition. A layout is the way in which the modified alerts are composed for further review by the analyst. In some embodiments the layout may be composed for presentation to an analyst, in different layouts, according to the analyst's role. In some embodiments the modified alert may be presented to the analyst in different windows or otherwise highlighted, according to the assigned priority.

The presentation logic receives the further modified alert to determine if the further modified alert is to be presented to an analyst for further review. The presentation logic may determine, based on the assigned priority of the further modified alert, to present the further modified alert to a cyber-security analyst. The presentation logic may determine, that a further modified alert shall not be presented to the cyber-security analyst due the relative priority (e.g., lesser) compared to other further modified alerts presented to the analyst at the same time. The relative priority of a further modified alert may increase (or decrease) based on selections made by a cyber-security analyst (e.g., as an analyst processes and addresses a first further modified alert, the relative priority of other further modified alerts may increase and be presented to the analyst).

The presentation logic may also process the course of action data included in the further modified alert to determine if a course of action may be automatically executed. A course of action to be automatically executed may be identified by the further modified alert. Automatic execution of the course of action may require communication with a conventional external computing device that is configured to effectuate the course of action (e.g., a firewall, switch, server

or endpoint system) connected to the network via the network interface. The mitigation logic receives a course of action for processing, the course of action may be received via the presentation logic if automatically selected or via an analyst interface when selected by an analyst. The mitigation logic initiates an external computing device (e.g., a cyber-security device, etc.) to execute a mitigation (i.e. via a course of action) sent by the mitigation logic.

More specifically, the mitigation logic processes the course of action received and launches processes based on the course of action. The executed course of action includes at least one process to be executed. Some processes to be executed as a course of action may require communication with one or more external computing devices through an interface (e.g., API calls to external computing devices, etc.). In some embodiments, courses of action may include more than one process, each process may be required by the course of action to be processed in series or parallel (in a temporally overlapping manner). A process may be required to be executed in series if the output of a first process is required as input of a subsequent process. If a process of the course of action executed does not process successfully, an alert may be generated by the mitigation logic and provided to the presentation logic for display to the cyber-security analyst. For example, a course of action may require a process A and a process B to operate in series. Process A may include the execution of an API call to a network connected firewall requesting the status of port **8081**, while Process B executes a process receiving the status, and if the status is "open", executes an API call to the network connected firewall to close port **8081**. Based on the success of the execution of the processes of the course of action, the mitigation logic communicates to the presentation logic. In some embodiments, the mitigation logic may provide an error message to the presentation logic, describing the nature of the failure if the course of action did not successfully complete. The meta-information associated with the processing by the mitigation logic (e.g., error messages, process success or failure, course of action success or failure, etc.) is provided in the form of an execution message. The mitigation logic may be configured to automatically, manually, or semi-automatically process courses of action.

The presentation logic receives data associated with the processing of a course of action by the mitigation logic (i.e. an execution message), via the mitigation logic. The data included in the received execution message is associated with the further modified alert and a determination is made by the presentation logic to present to an analyst. For example, the analyst may be provided a notification of a successful (or failed) execution of a course of action. In some embodiments an analyst may be presented with an alert describing the failed execution of a course of action as well as the associated further modified alert. The presentation logic provides the further modified alert to the storage logic for further processing.

The storage logic receives the further modified alert, from the presentation logic, and the associated execution message, and determines if the content received (e.g., the data associated with the further modified alert obtained from the execution message) should be stored in the knowledge store. The further modified alert may contain information about selections and results of course of action selected by an analyst and/or automatically selected by the presentation logic. The storage logic may parse the further modified alert to extract the selection of a course of action by an analyst to store in the knowledge store. In some embodiments, the storage logic may determine that a selected course of action

need not be stored in the knowledge store based on the success and/or failure of the course of action. In some other embodiments an execution message may be received directly from the mitigation logic, instead of being received via the presentation logic. Once processed by the storage logic, the presentation alert is provided to the reporting engine for display to the analyst.

The reporting logic is configured to provide reports via an interface to an analyst and/or a system administrator. The reporting logic may provide reports via an analyst interface and/or a network interface. The reporting logic generates the report for the analyst based on information provided by a received further modified alert. The reporting logic may be configured to generate discrete reports and/or dynamic interfaces for interaction by an analyst. The further modified alert to be displayed by the reporting interface, in combination with the system interface, may be displayed in addition to other further modified alerts that have been received by a dynamic interface. The analyst may interact with each further modified alert for analysis of the alert using additional information provided by the system and/or to select a course of action (which may also be included in the further modified alert). The interaction with the further modified alert may be received by an interface (e.g., a network interface and/or the analyst interface). The information received by the interface may be provided to the knowledge store via the storage logic. The information stored in the knowledge store is used by the predictive machine learning logic to generate a predictive model to implement a self-learning feedback loop. The self-learning feedback loop aids an analyst in efficiently addressing cyber-security alerts received by a cyber-security automated analyst alerting system.

Elements of the invention employ computerized techniques to generate machine learning models used to classify received alerts, initiate the display of classified received alerts, and re-generate the machine learning models in response to input received from a cyber-security analyst responsive to the displayed classified received alert.

I. Terminology

In the following description, certain terminology is used to describe features of the invention. For example, in certain situations, both terms "logic" and "engine" are representative of hardware, firmware and/or software that is configured to perform one or more functions. As hardware, logic (or engine) may include circuitry having data processing or storage functionality. Examples of such circuitry may include, but is not limited or restricted to a microprocessor, one or more processor cores, a programmable gate array, a microcontroller, an application specific integrated circuit, wireless receiver, transmitter and/or transceiver circuitry, semiconductor memory, or combinatorial logic.

Logic (or engine) may be software in the form of one or more software modules, such as executable code in the form of an executable application, an application programming interface (API), a subroutine, a function, a procedure, an applet, a servlet, a routine, source code, object code, a shared library/dynamic load library, or one or more instructions. These software modules may be stored in any type of a suitable non-transitory storage medium, or transitory storage medium (e.g., electrical, optical, acoustical or other form of propagated signals such as carrier waves, infrared signals, or digital signals). Examples of non-transitory storage medium may include, but are not limited or restricted to a programmable circuit; a semiconductor memory; non-persistent stor-

age such as volatile memory (e.g., any type of random access memory “RAM”); persistent storage such as non-volatile memory (e.g., read-only memory “ROM”, power-backed RAM, flash memory, phase-change memory, etc.), a solid-state drive, hard disk drive, an optical disc drive, or a portable memory device. As firmware, the executable code is stored in persistent storage. The term “computerized” generally represents that any corresponding operations are conducted by hardware in combination with software and/or firmware.

The term “transmission medium” (or “transmission media”) may refer to a communication path between two or more systems (e.g. any electronic devices with data processing functionality such as, for example, a security appliance, server, mainframe, computer, netbook, tablet, smart phone, router, switch, bridge or router). The communication path may include wired and/or wireless segments. Examples of wired and/or wireless segments include electrical wiring, optical fiber, cable, bus trace, or a wireless channel using infrared, radio frequency (RF), or any other wired/wireless signaling mechanism.

The term “alert” may refer to a signal or notification (e.g., report) received from, or issued by, a source. The alert conveys information regarding an event. An event may refer to an observed (or in some cases, inferred) occurrence that has significance to an associated alert type. An alert type may indicate an alert classification (e.g., an alert indicating a user login attempt may be classified as a “user alert”—i.e. an alert with a “user” type). A cyber-security event may be relevant to a cyber-threat. Relationships between events may be determined based on information provided by received cyber-security alerts describing events monitored by the cyber-security devices (or software). For example, a user-operated endpoint may be monitored by resident cyber-security software (e.g., an embedded agent), the software monitoring the execution of a process “opening” a file. An alert may be associated with, or triggered by, any of a variety of computing activities, for example: a granting or denial of administrative rights or escalation of privileges, an unauthorized access of an access-restricted compute device, detection of a new device on a restricted network, multiple different user login(s) made by a single compute device, an unexpected/unusual login of a user, detection of an internal vulnerability, etc.

The term “message” generally refers to signaling (wired or wireless) as either information placed in a prescribed format and transmitted in accordance with a suitable delivery protocol or information made accessible through a logical data structure such as an API. Hence, each message may be in the form of one or more packets, frame, or any other series of bits having the prescribed, structured format.

The term “object” generally refers to a collection of data, such as a group of related packets associated with a request-response message pairing for example, normally having a logical structure or organization that enables classification for purposes of analysis. For instance, an object may be a self-contained element, where different types of such objects may include an executable file, non-executable file (such as a document or a dynamically link library), a Portable Document Format (PDF) file, a JavaScript file, Zip file, a Flash file, a document (for example, a Microsoft Office® document), an electronic mail (email), downloaded web page, an instant messaging element in accordance with Session Initiation Protocol (SIP) or another messaging protocol, or the like.

The term “appliance” refers to any type of general-purpose or special-purpose computer, including a dedicated

computing device, adapted to implement any variety of existing, or future, software architectures relating to detection of, and protection from, cyberattack and related functionality. The term appliance should therefore be taken broadly to include such arrangements, in addition to any systems or subsystems configured to support such functionality, whether implemented in one or more network computing devices or other electronic devices, equipment, systems or subsystems.

The terms “computer”, “processor”, “computer processor”, “compute device”, or the like should be expansively construed to cover any kind of electronic device with data processing capabilities including, by way of non-limiting example, a digital signal processor (DSP), a microcontroller, a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a graphics processing unit (GPU), or any other electronic computing device comprising one or more processors of any kind, or any combination thereof.

As used herein, the phrase “for example,” “such as”, “for instance”, and variants thereof describe non-limiting embodiments of the presently disclosed subject matter. Reference in the specification to “one case”, “some cases”, “other cases”, or variants thereof means that a particular feature, structure or characteristic described in connection with the embodiment(s) is included in at least one embodiment of the presently disclosed subject matter. Thus the appearance of the phrase “one case”, “some cases”, “other cases” or variants thereof does not necessarily refer to the same embodiment(s).

It is appreciated that, unless specifically stated otherwise, certain features of the presently disclosed subject matter, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the presently disclosed subject matter, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination.

Lastly, the terms “or” and “and/or” as used herein are to be interpreted as inclusive or meaning any one or combination. Therefore, “A, B or C” or “A, B and/or C” mean “any of the following: A; B; C; A and B; A and C; B and C; A, B and C.” An exception to this definition will occur only when a combination of elements, functions, steps or acts are in some way inherently mutually exclusive.

II. Architecture

Referring to FIG. 1, an exemplary block diagram of an automated analyst alerting system **100** is communicatively coupled, via a network interface **110**, to at least one communication network **105**. The communication network **105** may couple the automated analyst alerting system **100** with cyber-security devices **102** and/or a remote analyst console **197** via transmission media to exchange information with the communication network directly or via the Internet. The communication network **105** may be coupled directly or indirectly to cyber-security device(s) **102**. The cyber-security devices **102** may operate within the same or different networks. Each cyber-security device represents a logical entity, operating on objects, to determine if they represent a cyber-security risk. In some embodiments a cyber-security device **102** may include a software application operating on a user operated endpoint device (e.g., a laptop, mobile phone, etc.) while in some other embodiments the cyber-security device may include a dedicated cyber-security appliance. The cyber-security device **102** may detect poten-

tial cyber-security threats and generate and issue a cyber-security alert. The cyber-security device **102** may be configured to direct issued alerts to the automated analyst alerting system **100**.

The automated analyst alerting system **100** includes a network interface **110**, an alert parser **120**, an alert recommendation engine **130**, a knowledge store **140**, a predictive model generation logic **150**, a presentation logic **160**, a mitigation logic **170**, a storage logic **180** a reporting engine **190** and an analyst interface **195**. Upon receipt by the automated analyst alerting system **100** of an alert generated by a cyber-security device **102**, via the network interface **110**, the alert is provided to the alert parser **120**. The alert parser **120** analyzes the received alert and normalizes the contents according to a set of normalization rules that normalize the received alert into a known alert format, comprehensible by the alert recommendation engine **130**. In some embodiments the normalization rules may be user-defined (and/or user-modifiable). In some embodiments the alert parser may be updated with additional (modified) processing (normalizing) rules. Such updates may be received by the automated analyst alerting system **100** periodically or aperiodically via the network interface **110**. The rule update may be processed by the alert parser **120** directly or via a separate logic (not shown).

The alert parser **120** provides the normalized alert to the alert recommendation engine **130** for further analysis. In some embodiments, the alert parser **120** may, limit further analysis of an alert based on contextual information. If a received alert received by the alert parser **120** includes a classification of the alert the alert parser may provide the alert recommendation engine **130** with the received alert classification and the alert recommendation engine **130** may include this classification (in some embodiments this classification may be added to the modified received alert without a confidence level). Contextual information may include data with respect to available system resources (e.g., processor load, memory availability, etc.), quality of alerts from particular cyber-security devices **120** (e.g., information related to reliability of cyber-security alerts in identifying cyberthreats associated with a particular cyber-security device), duplication (i.e. information that associates a set of alerts identifying identical alerts from cyber-security devices and associates them together for de-duplication by the various logics of the automated analyst alerting system), etc. Analysis of contextual information may be performed by the alert parser **120** by evaluating normalization rules by the alert parser **120**. By reducing the number of received alerts to be analyzed by the automated analyst alerting system **100**, the system may aid an analyst focus on high value alerts.

The alert recommendation engine **130** includes at least an alert analysis and labeling engine **132** and an action generator **134**. The alert recommendation engine **130** receives, from the alert parser **120**, an alert transformed according to the normalization rules and via the alert analysis and labeling engine **132**, generates at least one label associated with the alert as well as a confidence level associated with each label. The action generator **134** of the alert recommendation engine **130** receives the label and associated confidence levels from the alert analysis and labeling engine **132** and determines if an action may be associated with the alert. The components of the alert recommendation engine **130** (i.e. the alert analysis and labeling engine **132** and the action generator **134**) operate in concert with information provided by the knowledge store **140**.

The knowledge store **140**, operating in concert with the alert recommendation engine **130**, provides information

generated from the predictive model generation logic **150** and information received from analyst operation. The information provided to the knowledge store **140** may include, by non-limiting example, information associated with execution of mitigations by cyber-security device(s) **102**, information associated with the result of instructed mitigations by cyber-security device(s), classification of a received alert by an analyst, etc. Additionally, in some embodiments, the knowledge store **140** may include the predictive model generated by the predictive model generation logic **150**. In some embodiments the predictive model may be stored in a separate store (e.g., a store provided by the alert recommendation engine **130**, etc.). In some embodiments, the knowledge store **140** may receive and store information, from the analyst, associated with a classification of a received alert (e.g., related alerts, identifiers associated with the alert, intelligence associated with a received alert, etc.).

The predictive model generation logic **150** may periodically or aperiodically generate a predictive model to be used by the alert recommendation engine **130**. The predictive model generation logic may generate the predictive model in response to the receipt of a signal indicating new information has been stored in the knowledge store **140**. In some embodiments, the predictive model generation logic **150** may only generate a new model in response to the receipt by the knowledge store **140** of information received from an analyst (e.g., a new alert classification, a modification and/or update to an existing classification, correction of a previously mis-classified alert, etc.). The predictive model generated by the predictive machine learning model **150** may be generated according to a known machine learning recommendation techniques. In some embodiments the predictive machine learning logic **150** may train a predictive model based on the labelled data stored in the knowledge store **140**. In some embodiments, the predictive machine learning logic **150** may generate the predictive machine learning model "offline" (i.e., "out of band"). In some embodiments (not shown) the predictive machine learning logic **150** may be remotely located from the automated analyst alert system **100** and communicatively coupled, for example, via communication network **105**, with cloud computing resources (not shown). The generated predictive model generates at least one classification and/or association of the classification with an alert. In some embodiments the classification generated by the predictive model may correspond to a numerical association with the classification. For example, based on analysis of the alert by the predictive model generated by the predictive model generation logic **150**, an alert may be associated with (a) maliciousness (31% confidence level), (b) phishing (51% confidence level), and (c) benign (67% confidence level).

In some embodiments, the predictive model generation logic **150** may generate a predictive model associating mitigation actions ("actions") with identified classifications. In other embodiments, a separate logic (e.g., the action generator **134**) may determine an action associated with identified classifications. A set of known actions may be stored in the knowledge store **140**. In some embodiments, the analyst may generate (i.e. user-define) an action to be stored in the knowledge store. In certain embodiments, actions generated by an analyst, in response to an alert are automatically stored in the knowledge store **140**.

The alert analysis and labeling engine **132** receives from the alert parser **120** the received alert for further analysis and obtains a predictive model from the knowledge store **140**. The alert analysis and labeling engine **132** is configured to apply the obtained predictive model and apply the predictive

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model to the received alert. By applying the predictive model to the received alert the alert analysis and labeling engine **132** generates at least one classification label and confidence level. If a plurality of classification labels and levels of association of classifications are generated, the alert analysis and labeling engine **132** will determine a classification for the received alert. In some embodiments the alert analysis and labeling engine **132** may apply more than one classification to an alert. In some embodiments the classification determination of the alert analysis and labeling engine **132** may, by way of non-limiting example, include the classification corresponding to the highest confidence level, each classification where an associated level of classification exceeds a threshold, a classification associated with a level of classification exceeding a second threshold, higher than a first threshold, etc. In some embodiments the alert analysis and labeling engine **132** may provide the classification of the alert and the alert to the action generator **134** while in other embodiments, the alert analysis and labeling engine may provide the classification and the received alert directly to the presentation logic **160**.

The action generator **134** is configured to analyze the received alerts and classification to determine if a known action may be recommended to a receiving analyst. In some embodiments, the predictive model generation logic **150** may generate a predictive action model, stored in the knowledge store **140**. The predictive action model is adapted to, in combination with the action generator **134**, associate a known action with a received alert. In other embodiments the action generator may be configured with a set of rules associating specified actions with selected alerts. For example, an alert received and classified by the alert analysis and labeling engine **132** as being associated with “phishing” may cause the action generator **134** to associate an action to the alert, the action, upon execution, quarantines the cyber-security device **102** associated with the alert (i.e. the computing device associated with the phishing alert). Rules to be processed by the action generator **134** may be factor-set, and/or user (e.g., security administrator, analyst, etc.) configurable. The action generator may rely on information processed by the alert parser **120** associated with affected devices protected by the automated analyst alerting system **100**. In some embodiments the action generator **134** may identify an action associated with the alert to be automatically executed (e.g., not require execution approval from analyst). The action generator **134** may determine that no known (e.g., in the knowledge store **140**, and/or in configured rules of the action generator) action may be associated with the received alert and classification. Once an alert is analyzed by the action generator **134**, the alert is provided to the presentation logic **160**.

The presentation logic **160** receives, from the alert recommendation engine **130**, the received alert and associated classifications and actions. The presentation logic **160** determines if an associated action should be provided directly to the mitigation logic **170** or be presented to an analyst for determination. The presentation logic **160** may be configured to determine if and how an alert should be presented to an analyst by the reporting engine **190**. The presentation logic **160** may determine an alert whose associated action is to be automatically executed by the mitigation logic **170** should be presented to the analyst despite its automated execution. In some embodiments the presentation logic **160** may generate a graphical user interface (GUI) for the reporting engine **190** to present to the analyst. The presentation logic **160** may receive results associated with the execution of an action by the mitigation logic **170** and/or

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instructions received from the analyst related to alerts that were presented to the analyst. The presentation logic **160** provides the received alert and associated results and/or analyst instruction to the storage logic **180**.

The storage logic **180** determines if a received action, alert classification, or analyst instruction (e.g., action instruction, creation of a new action, etc.) should be stored in the knowledge store **140**. The determination as to whether an action should be stored in the knowledge store **140** may be based on whether the action is duplicative (e.g., a similar action exists in the knowledge store), not in opposition to existing actions stored in the knowledge store, etc. In some embodiments, a modification to an existing action may be received by the storage logic **180** and the contents of the knowledge store **140** may be modified. If no action needs to be stored in the knowledge store **140** or if it has already been stored in the knowledge store, the received information is provided to the reporting engine **190** for presentation to the analyst.

The mitigation logic **170** receives from the presentation logic **160** actions for execution by cyber-security device(s) **102**. The action generator **134** may identify, to the presentation logic **160** whether an action associated with an alert should be automatically executed by the mitigation logic. Similarly, the mitigation logic **170** may receive, via the network interface(s) **110**, an action instruction from an analyst (e.g. via the analyst interface **195**). The action instructed by the analyst to the mitigation logic **170** may be provided to the presentation logic **160** for further processing (as described above) and be further processed by the mitigation logic **170** for execution. The execution of actions by the mitigation logic **170** may be direct (e.g., an action which may be executed directly by the automated analyst system **100**) or indirect (e.g., issuing instructions, via the network interface(s) **110** to cyber-security device(s) **102**). In some embodiments the mitigation logic **170** may be configured with credentials for interaction with systems requiring authorization for executing cyber-security actions. The mitigation logic **170** may be configured to generate application programming interface (API) calls to cyber-security device(s) **102** in response to receiving an action for execution. In other embodiments an action may include the execution details and the mitigation logic **170** does not generate API calls to the cyber-security device(s) **102**. The result of an execution is received by the mitigation logic **170** via the network interface(s) **110** and provided to the presentation logic **160**.

The reporting engine **190** may be configured to generate an alert for transmission to an external display of an analyst. The reporting engine **190** may be configured to provide a GUI to the analyst display and/or other known display systems (e.g., command line terminal, etc.). The reporting engine **190** is configured to provide reports via the network interface(s) **110**, for example, the remote analyst console **197**. In some embodiments the reporting engine **190** may provide interactive alert which may allow an analyst to provide responsive instructions to the mitigation logic **170** for further processing by the automated analyst alerting system **100**. The analyst may provide an interactive response and consume alerts via the remote analyst console **197**.

As illustrated in FIG. 2 in greater detail, the automated analyst recommendation system **200** has physical hardware including hardware processors **210**, network interface(s) **220**, a memory **230**, a system interconnect **270**, and optionally, a user interface **290**. The memory **230** may contain software comprising an alert parser **240**, an alert analysis and labeling engine **242**, an action generator **244**, presenta-

tion logic 250, a mitigation logic 252, a reporting engine 254, an storage logic 260, and a predictive model generation logic 265. The physical hardware (e.g. hardware processors 210, network interface(s) 220, memory 230) may be connected for communication by the system interconnect 270, such as a bus. Generally speaking, an automated analyst recommendation system 200 is a network-connected alert analysis system configured to enhance the operation of a security operations center (SOC) by providing a SOC analyst with relevant alerts and meta-information.

The hardware processor 210 is a multipurpose, programmable device that accepts digital data as input, processes the input data according to instructions stored in its memory, and provides results as output. One example of the hardware processor 210 is an Intel® microprocessor with its associated instruction set architecture, which is used as a central processing unit (CPU) of the automated analyst recommendation system 200. Alternatively, the hardware processor 210 may include another type of CPU, a digital signal processor (DSP), an application specific integrated circuit (ASIC), or the like.

The network device(s) 280 may include various input/output (I/O) or peripheral devices, such as a storage device, for example. One type of storage device may include a solid state drive (SSD) embodied as a flash storage device or other non-volatile, solid-state electronic device (e.g., drives based on storage class memory components). Another type of storage device may include a hard disk drive (HDD). Each network device 280 may include one or more network ports containing the mechanical, electrical and/or signaling circuitry needed to connect the automated analyst recommendation system 200 to the private network 120 to thereby facilitate communications over the communication network 105. To that end, the network interface(s) 220 may be configured to transmit and/or receive messages using a variety of communication protocols including, inter alia, TCP/IP and HTTPS.

The memory 230 may include a plurality of locations that are addressable by the hardware processor 210 and the network interface(s) 220 for storing software (including software applications) and data structures associated with such software. The hardware processor 210 is adapted to manipulate the stored data structures as well as execute the stored software, which includes an alert parser 240, an alert analysis and labeling engine 242, an action generator 244, presentation logic 250, a mitigation logic 252, a reporting engine 254, an storage logic 260, and a predictive model generation logic 265.

The alert parser 240 is a software application, operating on data (i.e. alerts) provided to the automated analyst recommendation system 200 via the network interface(s) 220 according to the description of alert parser 120 of FIG. 1. The alert parser 240 receives an alert and processes the alert according a set of normalization rules residing within the memory 230. The alerts processed by the alert parser 240 are provided to the alert analysis and labeling engine 242 for further processing.

The alert analysis and labeling engine 242 processes received alerts according to a generated predictive model stored in memory 230. The alert analysis and labeling engine generates a set of classifications in response to the processing of the received alert by the predictive model. The classifications may correspond to a set of labels applied to the received alert and to be used in further processing of the alert by other components of the automated analyst recommendation system 200. The classification labels generated by the alert analysis and labeling engine 242 may include a

likelihood of association (i.e. confidence level) with the alert. The likelihood of association may be applied to the alert and provided, in addition to the associated classification label and alert, to the action generator 244. In some embodiments the alert analysis and labeling engine 242 may also generate a set of associated alerts related to the received alert. The association may result from the predictive model and/or be associated with correlating meta-information of the alert. The predictive model is generated by the predictive model generation logic 265.

The predictive model generation logic 265 generates predictive models and stores in the memory 230. In some embodiments the predictive model generation logic 265 may generate a separate second predictive action model (based on the actions previously associated with alerts and stored in the knowledge store 140) for use by the action generator 244, distinct and trained separately from the predictive model used by the alert analysis and labeling engine 242 (based on prior classifications of alerts and stored in the knowledge store 140). In other embodiments the predictive model generation logic may associate prior analyzed alerts with the received alert to determine if they are related and may need to be processed by the analyst together. If so, they may be associated together in meta-information and provided to the presentation logic 250. The predictive model generation logic 265 generates models based on information stored in memory 230 related to prior alerts and actions. The predictive model generation logic 265 analyzes stored information to generate a predictive model according to known machine learning techniques. A random forest classifier is an exemplary technique that creates a set of decision trees from randomly selected subset of training set. The random forest classifier then aggregates the decisions from the set of decision trees to decide the final classification associated with the targeted alert. In some embodiments an alternative technique may be used (e.g., convolutional neural networks, support vector machines, etc.). The generated predictive models are stored in memory 230 to be accessed by the analytic logics of the automated analyst recommendation system 200.

The action generator 244 receives from the alert analysis and labeling engine 242 the received alert and at least the classification label(s) determined by the alert analysis and labeling engine. The action generator 244 analyzes the received alert and classification and may determine an action which may be executed in response to the alert. The determined action may be an action recommended (to the analyst) to mitigate the cyber-security threat identified by the alert. In some embodiments the determined action may include instructions to obtain additional information regarding the alert (e.g., an instruction to the alert originating cyber-security device for additional meta-information related to the first alert). The action generator 244 may generate an action based on rules stored in memory 230 and/or based on model provided by the predictive model generation logic 265. The predictive model generation logic 265 may generate a predictive action model in response to storage in memory 230 of new actions. New actions may be stored in memory 230 based on an update action received by the automated analyst recommendation engine via the network interface(s) 220 and/or via analyst selecting a recommended action or submitting an action. The predictive action model is generated based on actions stored in memory 230. The action generator 244 may associate no actions or one or more actions in response to further analysis of the received alert and/or classification information (the classification information including the at least classification label and

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associated likelihood of association). In some embodiments the action generator **244** determines that a recommended action shall be executed without confirmation by the analyst and the action is labelled with such an indicator. Once the action generator **244** determines whether an action may be associated with the alert, the alert and any associated information is provided to the presentation logic **250**.

The presentation logic **250** is provided with the alert from either the alert analysis and labeling engine **242** or the action generator **244** as well as with additional meta-information (e.g., recommended action(s), classification(s) and associated confidence levels) generated during prior processing for presentation to the analyst. If an action is labeled for automatic execution the action is provided to the mitigation logic **252** by the presentation logic **250**. Similarly, if responsive to presentation to an analyst, the presentation logic **250** receives instructions from the analyst, the action instructed is provided to the mitigation logic **252** for processing. The presentation logic **250** may further analyze the alert and associated meta-information to determine a priority and arrangement of the alert and associated information to the analyst. For example, alerts associated with low confidence levels (e.g., the system cannot properly label the alert), may be assigned a higher priority and presented to the analyst. In some other embodiments, analysis of the meta-information associated with an alert may indicate duplicative alerts having been received, consequently, the presentation logic may generate a modified GUI to aggregate and/or filter the duplicative alerts to the analyst. In still yet other embodiments the presentation logic **250** may receive from the mitigation logic **252** the results of an executed action for presentation to the analyst and storage by the action logic **260**. Upon receipt, the execution results are associated with the associated alert's meta-information and provided to storage logic **260**.

The mitigation logic **252** receives action instructions via the presentation logic **250**. Actions may be provided to the mitigation logic **252** automatically or in response to an instruction from an analyst. The action may require communication via the network interface(s) **220** to third party systems (e.g., cyber-security devices **102**). Communication with third party systems may require authentication credentials for authorization, which may be configured by the security administrator and/or an analyst in advance of action execution or as needed. The mitigation logic **252** may also operate via the analyst alert recommendation system **200** directly. An action execution result may be generated upon receipt of results from an execution. In some embodiments, if no result response is received within a specified time period (e.g., 60 seconds) the mitigation logic may generate an action execution result indicating a "timeout". The results response is provided to the storage logic **260** via the presentation logic **250**.

The storage logic **260** processes the received alert and meta-information (including results information provided by the mitigation logic **252**). The storage logic **260** analyzes the alert and associated meta-information and determines if the action and/or classifications may be stored in memory **230**. The determination, as to whether or not the meta-information may be stored in memory **230**, may be based on the duplicative nature of the meta-information (i.e. determine if the same information is stored in the memory), modification of existing meta-information stored in the memory and/or if the meta-information to be stored is inconsistent with prior stored meta-information.

The reporting engine **254** receives the alert and associated meta-information for presentation to the analyst. The report-

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ing engine may provide the alert and associated meta-information to the user interface **290** and/or to the network device(s) **220** for presentation to the analyst. The user interface **290** may produce a graphical or textual based representation to a user of the endpoint **10** device **200**. The user interface **290** provides the user with the ability to interact with the computer. The user interface **290** may not be present for an endpoint device that is not dedicated to a single user or does not require the interaction with a user. The user interface **290** may receive input via the network device(s) **280** which include various input/output devices.

FIG. 3 represents an exemplary flowchart of a computerized method **300** for operating an automated analyst recommendation system **100**. The exemplary method **300** starts at step **305** and proceeds to step **310** where the automated analyst recommendation system **100** receives an alert from cyber-security device(s) **102** transmitted over the communication network **105** via the network interface(s) **110**. During step **310**, the alert parser **120** processes the alert to generate processible meta-information for further analysis by subsequent analytics logics (e.g., the alert analysis and labeling engine **132**, the action generator **134**, etc.). Upon completion of processing by the alert parser **120**, the alert and associated meta-information is provided to alert analysis and labeling engine **132** for further analysis in step **315**.

The alert analysis and labeling engine **132**, during step **315** applies the predictive machine learning model stored in the knowledge store **140**, to the received alert and associated meta-information. The results of the analysis of the received alert and meta-information with the predictive model is at least one classification label and a confidence level (e.g., likelihood of association, etc.). In some embodiments the predictive model may also generate a set of alerts associated as meta-information with the received alert. The associated alerts may be relevant to the assessment of the received alert when reviewed by an analyst. If associated alerts are identified, the alert and associated meta-information is retrieved from the knowledge store **140** and added to the received alert's associated meta-information for further processing in step **320**. In step **325** the meta-information and the received alert are associated and provided to the action generator **134** for further analysis.

In step **330** the action generator **134** receives the alert received by the automated analyst recommendation system **100** and associated meta-information for analysis. The analysis may include the processing of factory-set and/or user-defined rules. For example, an alert associated with a "phishing" email cyberattack may identify the source cyber-security device(s) **102** from the meta-information and generate an action targeting the phishing email for quarantine. In some embodiments a predictive action model may be applied to the received alert and meta-information to generate a set of recommended actions based on prior actions taken and/or recorded by the automated analyst recommendation system **100**. If an action is generated in step **335**, the method continues step **340** where the alert and its associated meta-information is modified with the generated action(s). Further processing by the action generator **134** may further determine if at least one of the generated action(s) should be automatically processed by the mitigation logic **170** in step **345**. If the generated action is determined to be automatically run in step **350**, and succeeds, the method ends at step **390**. If the generated action is determined to be manually run in step **345**, the generated action is presented to the analyst via the analyst interface **195** in step **348**. Upon selection by the analyst, the analyst interface **195** provides the selection

to the mitigation logic for execution and if in step 350 the executed action succeeds, the method ends at step 390.

If the action generator cannot identify a recommendable action in step 335 or the executed action fails in step 350, the presentation logic 160 determines a priority for presentation of the alert to the analyst in step 360. The determination of priority is based, at least in part, on the success of an action executed by the mitigation logic 170. In some embodiments, the priority for presentation of an alert to the analyst may be based on the confidence level associated with a classification of the alert. In some embodiments, the presentation logic 160 determines a priority level of an alert in step 360 then in step 365 determines if the alert, based in part on the priority level, should be presented to the analyst. If the presentation logic 160 determines that the alert need not be presented to the analyst in step 370, the method ends at step 390.

If the alert is determined to be presented to the analyst in step 370 by the presentation logic 160, the alert is presented to the analyst for further interaction. In some embodiments the further interaction with the analyst may be done through a user interface 290 or via the reporting engine 190 once the alert has been further processed by the storage logic 180. In step 375 the analyst is presented with the modified alert. The analyst may select an action associated with the modified alert, modify a classification of the modified alert, and/or generate an action or classification associated with the alert based on the context received. The result of step 375 is provided to the knowledge store via the storage logic 180 in step 380. In step 380, upon receipt of a new and/or modified alert and/or action result, the storage logic 180 may store the received information in the knowledge store 140. If information received by the knowledge store in step 380, the alert analysis and labeling engine 132 may regenerate a new predictive model based on the new information and re-analyze the received alert in step 385. By this method, the system will identify alerts requiring additional action by an analyst while minimizing the time spent by analysts on low value alerts.

The foregoing description has been directed to specific embodiments. It will be apparent, however, that other variations and modifications may be made to the described embodiments, with the attainment of some or all their advantages. For instance, it is expressly contemplated that the components and/or elements described herein can be implemented as software encoded on a tangible (non-transitory) computer-readable medium (e.g., disks, electronic memory, and/or CDs) having program instructions executing on a computer, hardware, firmware, or a combination thereof. Moreover, the embodiments or aspects thereof can be implemented in hardware, firmware, software, or a combination thereof. In the foregoing description, for example, in certain situations, terms such as “engine,” “component” and “logic” are representative of hardware, firmware and/or software that is configured to perform one or more functions. As hardware, engine (or component/logic) may include circuitry having data processing or storage functionality. Examples of such circuitry may include, but is not limited or restricted to a microprocessor, one or more processor cores, a programmable gate array, a microcontroller, an application specific integrated circuit, semiconductor memory, or combinatorial logic. Accordingly, this description is to be taken only by way of example and not to otherwise limit the scope of the embodiments herein. Therefore, it is the object of the appended claims to cover all such variations and modifications as come within the true spirit and scope of the invention.

What is claimed is:

1. A computer-implemented method to perform self-learning for a predictive machine learning model of a cyber-security alert system, the method comprising:
 - obtaining, by a computing system, the predictive machine learning model, the predictive machine learning model trained based on data in a knowledge store;
 - receiving, by the computing system, an alert associated with a monitored network;
 - classifying, by the computing system, the received alert according to the predictive machine learning model to generate at least one alert classification;
 - automatically generating, by the computing system, a one or more recommended actions responsive to, and associated with, the received alert based on the alert classification;
 - automatically causing, by the computing system, execution of the one or more recommended actions;
 - updating, by the computing system, the knowledge store to include a result of the one or more recommended actions in the knowledge store; and
 - generating an updated predictive machine learning model based on the updated knowledge store.
2. The computer-implemented method of claim 1, wherein the result indicates a success or failure of the one or more recommended actions.
3. The computer-implemented method of claim 1, further comprising surfacing, by the computing system, the at least one alert classification or the one or more recommended actions for modification by an analyst.
4. The computer-implemented method of claim 3, wherein said surfacing is performed in response to the result of the one or more recommended actions indicating a failure of the one or more recommended actions.
5. The computer-implemented method of claim 3, further comprising, prior to generating the updated predictive machine learning model: updating the knowledge store based on a modification entered by the analyst.
6. The computer-implemented method of claim 1, wherein the predictive machine learning model comprises an artificial neural network.
7. The computer-implemented method of claim 1, wherein the predictive machine learning model generates a confidence score for the at least one alert classification and wherein automatically causing, by the computing system, execution of the one or more recommended actions occurs in response to the confidence score exceeding a confidence threshold.
8. The computer-implemented method of claim 1, wherein the one or more recommended actions comprise communication with a conventional external computing device that is configured to effectuate the one or more recommended actions.
9. The computer-implemented method of claim 8, wherein said communication occurs via an Application Programming Interface (API) call.
10. A computing system configured to perform a self-learning loop for a predictive machine learning model of a cyber-security alert system, the computing system comprising:
 - one or more processors; and
 - one or more non-transitory computer-readable media that collectively store:
 - a knowledge store;
 - a predictive machine learning model; and

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instructions that, when executed by the one or more processors cause the computing system to perform operations, the operations comprising:

obtaining, by the computing system, the predictive machine learning model, the predictive machine learning model trained based on data in the knowledge store;

receiving, by the computing system, an alert associated with a monitored network;

classifying, by the computing system, the received alert according to the predictive machine learning model to generate at least one alert classification;

automatically generating, by the computing system, a one or more recommended actions responsive to, and associated with, the received alert based on the alert classification;

automatically causing, by the computing system, execution of the one or more recommended actions;

updating, by the computing system, the knowledge store to include a result of the one or more recommended actions in the knowledge store; and

generating an updated predictive machine learning model based on the updated knowledge store.

11. The computing system of claim 10, wherein the result indicates a success or failure of the one or more recommended actions.

12. The computing system of claim 10, further comprising surfacing, by the computing system, the at least one alert classification or the one or more recommended actions for modification by an analyst.

13. The computing system of claim 12, wherein said surfacing is performed in response to the result of the one or more recommended actions indicating a failure of the one or more recommended actions.

14. The computing system of claim 12, further comprising, prior to generating the updated predictive machine learning model: updating the knowledge store based on a modification entered by the analyst.

15. The computing system of claim 10, wherein the predictive machine learning model comprises an artificial neural network.

16. The computing system of claim 10, wherein the predictive machine learning model generates a confidence

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score for the at least one alert classification and wherein automatically causing, by the computing system, execution of the one or more recommended actions occurs in response to the confidence score exceeding a confidence threshold.

17. The computing system of claim 10, wherein the one or more recommended actions comprise communication with a conventional external computing device that is configured to effectuate the one or more recommended actions.

18. The computing system of claim 17, wherein said communication occurs via an Application Programming Interface (API) call.

19. The computing system of claim 10, wherein the operations further comprise employing the updated predictive machine learning model to process a new alert.

20. One or more non-transitory computer-readable media that collectively store:

a knowledge store;

a predictive machine learning model; and

instructions that, when executed by one or more processors of a computing system cause the computing system to perform operations, the operations comprising:

obtaining, by the computing system, the predictive machine learning model, the predictive machine learning model trained based on data in the knowledge store;

receiving, by the computing system, an alert associated with a monitored network;

classifying, by the computing system, the received alert to generate at least one alert classification;

automatically generating, by the computing system and using the predictive machine learning model, one or more recommended actions responsive to, and associated with, the received alert based on the alert classification;

automatically causing, by the computing system, execution of the one or more recommended actions;

updating, by the computing system, the knowledge store to include a result of the one or more recommended actions in the knowledge store; and

generating an updated predictive machine learning model based on the updated knowledge store.

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